

ML Assignment - 2: Laptop Price Predictor

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Course: Machine Learning

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1. GitHub Repository Link

>> <https://github.com/Rakeshraj06/ML-assingment-2-laptop-price-predictor->

Repository Contents:

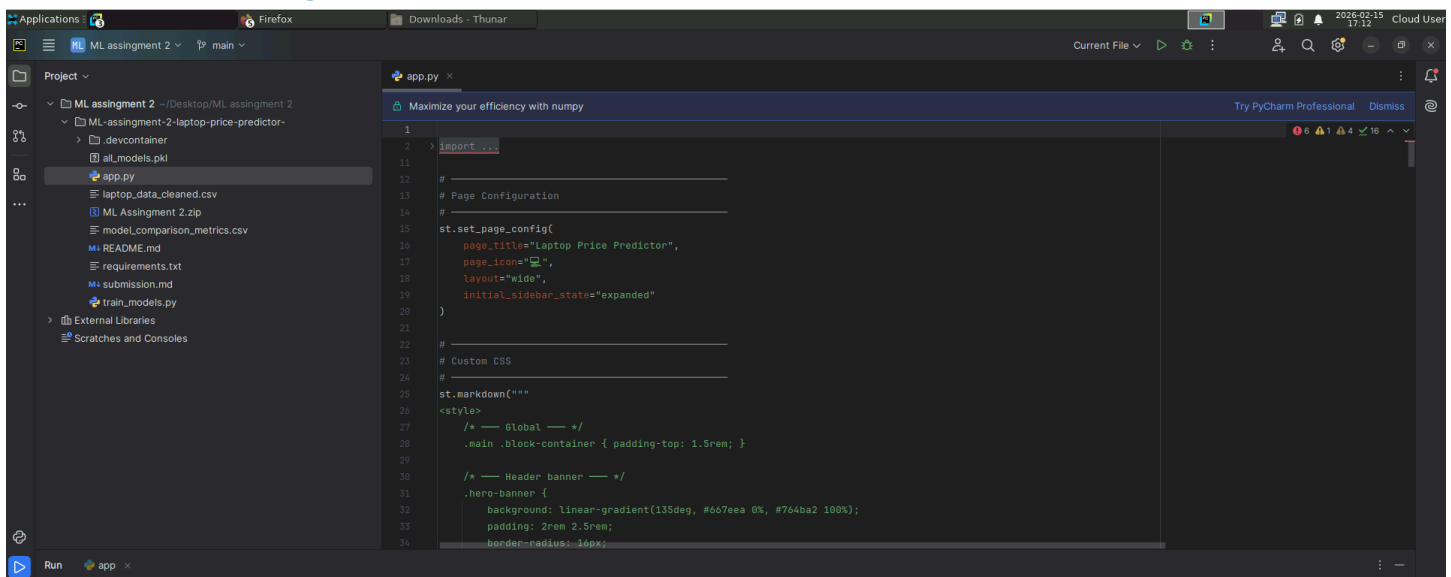
- Complete source code (app.py, train_models.py, etc.)
- Dependencies (requirements.txt)
- Comprehensive Project/Assignment Documentation (README.md)
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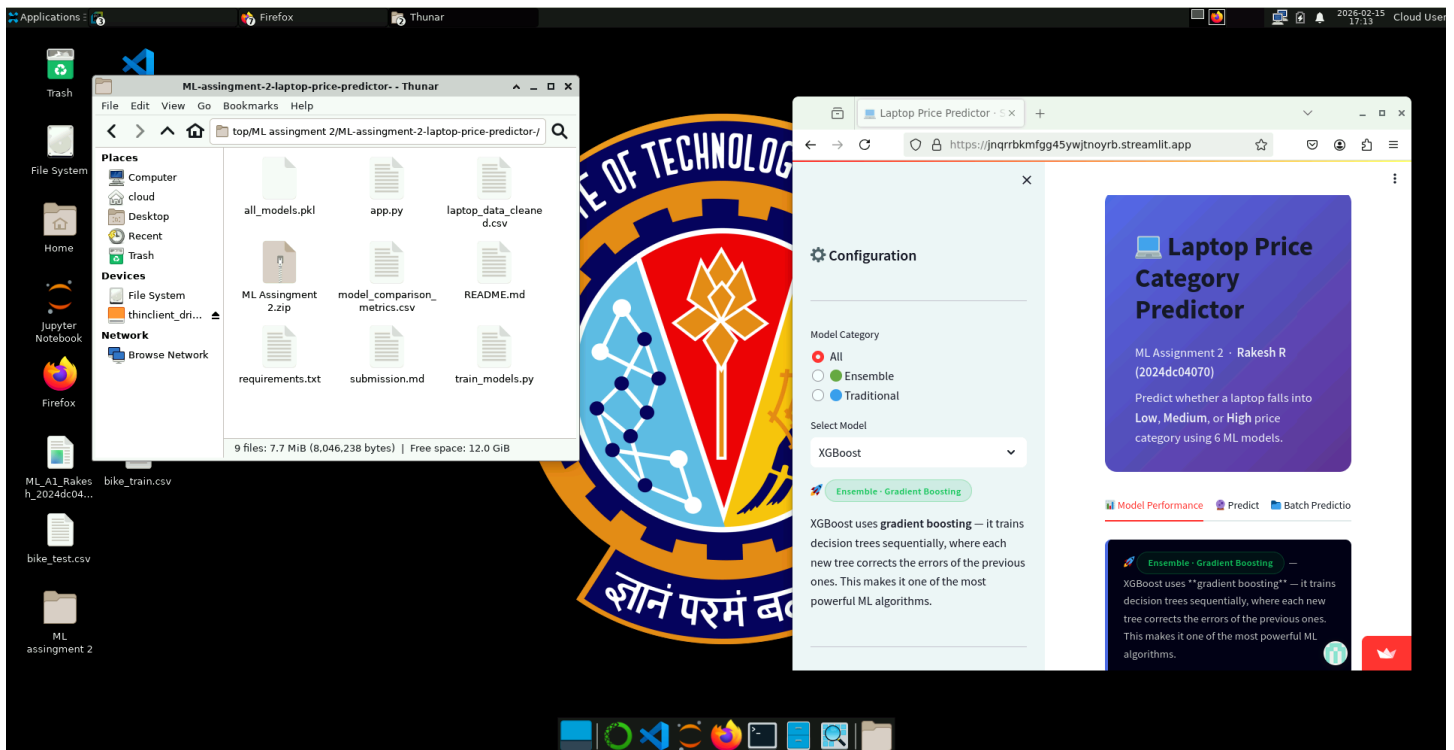
2. Live Streamlit App Link

>><https://jnqrrbkmfgg45ywjtnoyrb.streamlit.app/>

(Deployed using Streamlit Community Cloud — Interactive Frontend)

3. Screenshot of Assignment Execution (BITS Virtual Lab)





4. GitHub README Content

Laptop Price Classification & Prediction

Multi-class classification of laptops into Low, Medium, and High price categories using six machine learning models.

Section A: Problem Statement & Objectives

The goal of this project is to build and compare six machine learning classification algorithms to predict laptop price categories. The continuous price data was transformed into discrete categories (Low, Medium, High) using quantile-based binning to ensure balanced classes.

- Perform multi-class classification on structured laptop specifications.
- Compare traditional models (Logistic Regression, Decision Tree, KNN, Naive Bayes) against advanced ensemble methods (Random Forest, XGBoost).
- Deploy an interactive Streamlit dashboard for real-time predictions and model evaluation.

Section B: Dataset Description

Property	Detail
Source	Kaggle Dataset – laptop-price-prediction-cleaned-dataset
Instances	1,273
Features	12 (5 categorical, 7 numerical)
Target	Price_Category — Low · Medium · High
Split	80 / 20 stratified

Section C: Models Used & Comparative Analysis

ML Model Name	Accuracy	AUC Score	Precision	Recall	F1 Score	MCC
XGBoost (Ensemble)	0.8745	0.9421	0.8771	0.8745	0.8753	0.8122
Random Forest (Ensemble)	0.8471	0.9158	0.8538	0.8471	0.8491	0.7717
Logistic Regression	0.8078	0.8812	0.8237	0.8078	0.8116	0.7159
Decision Tree	0.8078	0.8545	0.8160	0.8078	0.8100	0.7133
KNN	0.8039	0.8421	0.8117	0.8039	0.8061	0.7073
Naive Bayes	0.5137	0.6542	0.6031	0.5137	0.4731	0.3064

Technical Observations

ML Model Name	Observation about model performance
Logistic Regression	Provides a strong baseline. Efficiently identifies linear relationships in hardware specs.
Decision Tree	Interpretable but less stable; accuracy is comparable to baseline methods.
kNN	Effective at local pattern matching based on similar laptop configurations.
Naive Bayes	Weakest performer; the independence assumption is violated by correlated features like RAM and CPU brand.
Random Forest (Ensemble)	Significant boost in stability and accuracy using the bagging technique.
XGBoost (Ensemble)	Top performer (87.5% Accuracy); gradient boosting effectively minimizes residual errors.

- Top Performer: XGBoost (87.45%) sequentially minimizes errors, making it highly effective for these hardware feature correlations.
- Robustness: Random Forest (84.71%) provides strong generalization through its bagging technique.
- Linearity: Logistic Regression (80.78%) shows hardware specs have a strong linear relationship with price buckets.
- Constraint: Naive Bayes (51.37%) fails to capture dependencies between components (e.g., RAM and GPU tier).

Streamlit App Features

- Project upload option (CSV): Batch prediction support for test data files.
- Model selection: Choose between 6 diverse ML models via dropdown.
- Evaluation metrics: Live display of Accuracy, AUC, and F1.
- Visual Evidence: Integrated Confusion Matrix (Heatmap) and Classification Report.

Tools Used

- Python
- Scikit-learn, XGBoost
- Streamlit

- Pandas, Plotly, Seaborn